Florva: An AI-Powered Gardening and Plant Care Platform

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1. Project Overview

Florva is a web-based gardening platform that allows customers to purchase gardening products, book gardening services, manage their plants, and receive assistance through an AI-powered chatbot. The system will initially support one gardener and an administrator.

2. User Roles

Customer

a. Register, log in, and manage their profile.
b. Browse gardening products and services.
c. Purchase products and make online payments.
d. Book gardening services.
e. Track orders and bookings.
f. Manage personal plants and receive care reminders.
g. Use the AI chatbot for plant care assistance.

Gardener

a. Manage gardening products, prices, descriptions, and stock.
b. Add and manage gardening services.
c. Set appointment availability.
d. Accept or reject booking requests.
e. View customer orders and update order statuses.
f. View sales summaries, booking statistics, and monthly performance charts.

Administrator

a. Manage customer and gardener accounts.
b. Monitor products, services, orders, and bookings.
c. Manage plant information and categories.
d. Monitor payment transactions.
e. View overall revenue, sales, booking statistics, customer activity, and monthly performance charts.

3. Functional Requirements

3.1 Authentication and User Management

a. Secure registration and login.
b. Role-based access for customers, gardeners, and administrators.
c. Profile management.

3.2 Gardening Product Management

a. Browse, search, and filter gardening products.
b. View product details, prices, images, and availability.
c. Add products to a shopping cart.
d. Allow the gardener to add, edit, and remove products.
e. Manage product stock and availability.

3.3 Gardening Services and Booking

a. Browse available gardening services and their pricing.
b. Select services and request appointments.
c. Choose available dates and time slots.
d. Allow the gardener to accept or reject booking requests.
e. View booking status and history.

3.4 Shopping Cart and Payment

a. Add, remove, and update cart items.
b. Calculate order totals.
c. Place orders and complete online payments through a sandbox payment gateway.
d. Maintain order and payment history.
e. Update and monitor order statuses.

3.5 Personal Plant Management

a. Add and manage personal plants.
b. Record plant names, planting dates, and notes.
c. Provide basic watering and fertilizing schedules.
d. Display in-app plant care reminders.

3.6 AI-Powered Plant Care Chatbot

a. Answer gardening and plant care questions.
b. Provide recommendations about sunlight, watering, soil, and fertilizers.
c. Suggest suitable plants based on customer preferences.
d. Provide guidance on common plant problems.
e. Integrate the Gemini API for AI-generated responses.
f. Maintain individual customer chatbot conversation histories.

3.7 Dashboard and Reports

a. Provide separate dashboards for the gardener and administrator.
b. Display sales, orders, bookings, and revenue summaries.
c. Show monthly sales and booking trends using charts and graphs.
d. Display popular products and services.
e. Allow the administrator to monitor overall platform activity.
f. Allow the gardener to monitor their own sales and booking activities.